

Problem Set 4: Due on Friday, March 6, @noon**Recursive Competitive Equilibria**

(100 points total)

1. **RCE with inelastic labor supply (10 points).** Consider an economy with a continuum of consumers of mass one. Each consumer's preferences are given by

$$E \sum_{t=0}^{\infty} \beta^t \theta_t u(c_t),$$

where $\beta \in (0, 1)$, c_t is consumption in period t , and θ_t is idiosyncratic taste shock. Each consumer faces a Markov process for the taste shock that has the transition function Q . The function Q is the same for all consumers, but the shock realizations are i.i.d. across consumers. Notice that consumers do not value leisure.

Consumers own capital which evolves according to the usual capital accumulation technology. Consumers supply capital and labor to the representative firm, which has the Cobb-Douglas production technology $Y = K^\alpha N^{1-\alpha}$.

- (a) (5 points) Define a recursive competitive equilibrium in this economy. Here and in all questions below, make sure to write an explicit expression for the law of motion for the probability distribution of consumers across individual states.
 - (b) (5 points) Show that the capital rental rate in this economy depends only on the aggregate capital, K . Given this fact, can one write a definition of a recursive competitive equilibrium with K as the only aggregate state variable? If your answer is “yes”, write down the definition, if your answer is “no”, carefully explain it.
2. **RCE with a fixed population of skilled and unskilled workers (10 points).** Consider the following optimal growth model with skilled and unskilled labor. There is a mass one of individuals, a constant fraction θ of which is skilled, and a fraction $1 - \theta$ is unskilled. Individuals within each group are identical. The skill type of an individual is fixed and does not change over time. Each individual is endowed with one unit of time and maximizes

$$\sum_{t=0}^{\infty} \beta^t U(c_{it}, 1 - n_{it}),$$

where c_{it} and n_{it} are consumption and hours worked in period t by an individual of type $i \in \{s, u\}$ (where s and u stand for the skilled and the unskilled). Both types accumulate capital according to the usual technology, and supply capital and labor services to the representative firm.

The only way in which individuals of the two types differ *ex-ante* is how their labor inputs enter into the firm's production technology. Specifically, the production technology is constant returns to scale in capital, skilled labor, and unskilled labor: $F(K, N_s, N_u)$. All individuals are endowed with the same initial level of capital, $k_0 > 0$. Assume for simplicity that there is no bond market.

- (a) (6 points) Define a recursive competitive equilibrium in this economy.
- (b) (4 points) Do equilibrium prices depend only on the average capital holdings, or the distribution of capital holdings? Explain your answer carefully.

3. **RCE with tradable vs. non-tradable home production good (30 points).** Consider an economy populated by a continuum of ex-ante identical households of measure one. There are two goods in the economy, 1 and 2. Each household maximizes

$$E \sum_{t=0}^{\infty} \beta^t U(c_{1,t}, c_{2,t}),$$

where $c_{1,t} \geq 0$ and $c_{2,t} \geq 0$ denote consumption of the two goods, the function U is strictly increasing in both arguments, strictly concave, differentiable, satisfies $\lim_{c_1 \rightarrow 0} U(c_1, c_2) = \lim_{c_2 \rightarrow 0} U(c_1, c_2) = +\infty$, and $\beta \in (0, 1)$.

Good 1 is produced by the representative firm from capital and labor using the production technology $F(K, N)$, where F is strictly increasing in both arguments, differentiable, strictly concave, and has constant returns to scale. Good 1 is traded on the market and is used for consumption and investment. Each household accumulates capital according to $k_{t+1} = (1 - \delta)k_t + i_t$, where i_t is investment in terms of good 1, and $\delta \in [0, 1]$.

Good 2 is produced at home by each household using *their own labor only* according to the linear technology θh , where h denotes the hours spent on production of good 2, and $\theta (> 0)$ is the idiosyncratic productivity of the household in producing this particular good. The random variable θ is i.i.d. over time and across households, distributed according to the probability measure ϕ , where ϕ is the same for all households. Just as good 1, good 2 is *traded on the market*.

Each household is endowed with an initial capital stock $k_0 > 0$ and one unit of time in each period, which it chooses to split between working for the firm (producing good 1) and working at home (producing good 2). Households do not have access to insurance markets, and cannot save/borrow.

- (a) (15 points) Define a recursive competitive equilibrium for this economy.
- (b) (5 points) Characterize the optimal decision of each household about how to split its time endowment between working for the firm and working at home. Explain.
- (c) (5 points) Suppose now that good 2 is non-tradable, so it has to be consumed entirely within the household that produced it. How does the household's problem change in

this case? How does the optimal labor choice change compared to the previous case? Explain.

- (d) (5 points) Economist X claims that while equilibrium in the original setup is Pareto efficient, the equilibrium in the case in which good 2 is non-tradable is not. Is s/he right?

4. **RCE with sector-specific human capital (25 points).** Consider an economy with two sectors, A and B . There are two representative firms, one in each sector. The firm in sector $i \in \{A, B\}$ operates a production technology $z_i f(H)$, where H denotes the efficiency units of labor hired, f is strictly increasing and strictly concave, and z_i is the sector-specific aggregate productivity. Normalize $z_B = 1$, and let $z \equiv z_A$ be the relative productivity of sector A . The relative productivity z follows a Markov process with the transition function Q . The firms hire efficiency units of labor at competitive wage rates.

There is a continuum of *ex-ante* identical consumers of measure 1. Each consumer maximizes his present discounted utility of lifetime consumption,

$$E \sum_{t=0}^{\infty} \beta^t u(c_{it}),$$

where $c_{it} \geq 0$ denotes consumption of household i at time t , u is strictly increasing and strictly concave, and $0 < \beta < 1$.

Each agent is endowed with one unit of time in each period that they can spend working in (only) one of the two sectors in a given period. Each consumer has sector-specific levels of human capital, denoted by h_A and h_B . If the consumer chooses to work in sector i in a particular period, then he supplies h_i efficiency units of labor to the firm in that sector. His sector- i -specific human capital next period is $(1 + \varepsilon)h_i$ if he worked in sector i this period, and is $(1 - \delta)h_i$ if he did not work in sector i this period, where $\varepsilon > 0$ and $\delta > 0$. At time zero, all consumers are endowed with the same level $\bar{h}$ of human capital in each sector: $h_{A,0} = h_{B,0} = \bar{h}$.

The consumers do not trade any assets. All consumers have equal ownership of each of the two firms.

Define a recursive competitive equilibrium for this economy.

5. **RCE with two Sectors and heterogeneous firms (25 points).** This question asks you to define a recursive competitive equilibrium in an economy with two sectors and idiosyncratic productivity shocks. Consider an economy populated by a large number of consumers and a continuum of firms of measure 1. There are two consumption goods that are produced in two different sectors, A and B . In each period, each firm can operate in one of the two sectors. Firms differ in their (relative) productivities. Specifically, a firm's productivity in sector A at date t , denoted by $z_{A,t}$, is a random variable, which is *i.i.d.* over time and across firms, and

can take S possible values, $\bar{z}_1, \dots, \bar{z}_S$ (where $\bar{z}_1 < \dots < \bar{z}_S$), with corresponding probabilities $\pi_1, \dots, \pi_S$. Productivity in sector B, $z_{B,t}$, is the same over time and across firms and is normalized to one. Production technology in sector $i \in \{A, B\}$ of a firm with productivity $z_{A,t}$ (and $z_{B,t} = 1$) is given by $z_{i,t}f(n)$, where n is the labor input, and the function f is strictly increasing and strictly concave.

In each period, after observing the realization of its productivity, each firm chooses which sector to operate in. If the firm chooses to *switch* the sector relative to previous period, it has to pay a constant adjustment cost $\psi \geq 0$. If the firm remains in the same sector as in the previous period, it pays no such cost. Upon deciding which sector to operate in, the firm decides how much labor to hire given a competitive wage.

Consumers are identical. Each consumer has preferences given by

$$\sum_{t=0}^{\infty} \beta^t u(c_{A,t}, c_{B,t}),$$

where $c_{A,t} \geq 0$ and $c_{B,t} \geq 0$ are consumption levels of the goods produced in sectors A and B , respectively, and $\beta \in (0, 1)$. (Notice that consumers do not value leisure.) Each consumer is endowed with one unit of time in each period that s/he can perfectly divide between working in either of the two sectors. Suppose for simplicity that consumers cannot (or choose not to) borrow or save, and firms discount future profits at the risk-free rate $r = \beta^{-1} - 1$. All consumers have equal ownership of the firms.

Define a recursive competitive equilibrium in this economy.